Reviewer Report: Burmese-to-English Machine Translation

Manuscript: Burmese_to_English_MT_Report.docx

Summary

This manuscript documents a five-phase study of neural machine translation for the Burmese-English language pair. The reported progression is: Phase 1 custom LSTM baseline (BLEU 2.05), Phase 2 optimized LSTM (BLEU 4.20), Phase 3 NLLB-200 zero-shot (BLEU 16.07), Phase 4 NLLB-200 with LoRA fine-tuning (BLEU 21.87), and Phase 5 planned OPUS-100 augmentation with an expected BLEU of 25-30+. The work follows a sensible engineering trajectory, but the manuscript lacks the experimental detail, statistical validation, and contextual comparison expected in a research publication. This review provides detailed feedback on each phase.

1. Detailed Experimental Feedback

1.1 Phase 1: Baseline LSTM (BLEU 2.05)

Experimental design The baseline uses a Sequence-to-Sequence LSTM with Luong attention, one LSTM layer, 100-dimensional embeddings, and a 256-dimensional hidden state. The model is trained for five epochs. The four limitations identified (random embeddings, 100% teacher forcing, shallow architecture, short training) are reasonable starting hypotheses.

Methodological concerns The most serious problem is that the manuscript does not describe how the Burmese text was prepared before tokenization. Burmese text frequently appears in legacy Zawgyi-One encoding, which is visually identical to Unicode but uses different code points. If the training data contains mixed Zawgyi and Unicode, the model's vocabulary is effectively doubled for the same characters. This is likely a major, unmeasured contributor to the BLEU 2.05 result. The manuscript should report the encoding distribution of the data and the normalization steps applied.

A second issue is segmentation. Burmese is a scriptio continua language with no word boundaries. The manuscript mentions SentencePiece BPE, but BPE on raw Burmese characters can produce subword units that split syllables in linguistically unnatural places. Established Myanmar-English NMT work has found that syllable-level segmentation before BPE improves results. The manuscript should state whether syllable segmentation was used and, if not, why.

A third issue is the vocabulary size. With only 100-dimensional embeddings and a single LSTM layer, the model has limited capacity, but the manuscript does not report the vocabulary size or the out-of-vocabulary rate on the test set. A BLEU of 2.05 could indicate that the model is simply predicting the most common English words because it does not understand most Burmese tokens.

Interpretation of BLEU 2.05 A BLEU score of 2.05 is effectively non-functional. It suggests that the model's output has almost no n-gram overlap with the reference translations. This is consistent with the identified limitations, but it also suggests that the baseline may have additional problems, such as poor data quality or incorrect tokenization, that are not discussed. The manuscript should verify that the baseline can at least copy or produce common phrases correctly.

Recommendations

- Document the Burmese text preprocessing pipeline: Zawgyi-to-Unicode conversion, Unicode NFC normalization, syllable segmentation if used, and BPE training details.
- Report the vocabulary size, OOV rate, and encoding distribution of the training and test data.
- Include sample translations from the baseline to illustrate the failure modes.
- Verify that the BLEU computation uses the same tokenization for the hypothesis and the reference.

1.2 Phase 2: Optimized LSTM (BLEU 4.20)

Experimental design Phase 2 applies four changes simultaneously: on-the-fly Word2Vec embeddings, 50% scheduled sampling, doubled embedding and hidden dimensions with a second LSTM layer, and extended training to 30 epochs. The reported BLEU improves from 2.05 to 4.20, a 104% relative increase.

Methodological concerns The four improvements are all technically reasonable, but they are not isolated. The manuscript does not report an ablation study, so it is impossible to determine which change drove the gain. For example:

- If Word2Vec embeddings account for most of the improvement, this would confirm that Phase 1 suffered mainly from poor initialization.
- If scheduled sampling accounts for most of the improvement, this would confirm that exposure bias was the dominant problem.
- If model capacity accounts for most of the improvement, this would suggest that the baseline was structurally too small.

Without an ablation table, the Phase 2 result is difficult to interpret. The manuscript should report the BLEU score after each incremental change.

Another concern is the absolute value. A BLEU of 4.20 is still far below practical usability. The manuscript correctly notes this, but it should also explain why LSTMs are structurally limited for Burmese-English. Burmese is an SOV language with sentence-final verbs, while English is SVO. An LSTM must carry information from the beginning of the sentence to the end before it can generate the English verb in the correct position. A shallow LSTM with 256 hidden units is unlikely to do this reliably. This point should be discussed.

Training dynamics The manuscript reports 30 training epochs but does not show training or validation loss curves. Without these curves, it is impossible to know whether the model has converged or overfitted. The 30-epoch choice should be justified by showing that validation BLEU plateaus around that point.

Recommendations

- Provide an ablation study that adds one improvement at a time and reports BLEU after each addition.
- Include training and validation loss curves as well as validation BLEU per epoch.
- Analyze sample translations to identify whether the remaining errors are due to vocabulary, word order, or fluency.
- Discuss the SOV-to-SVO reordering problem as a structural reason for the LSTM ceiling.

1.3 Phase 3: NLLB-200 Zero-Shot (BLEU 16.07)

Experimental design The authors switch to the NLLB-200-Distilled-600M model and evaluate it without any fine-tuning on the Burmese-English data. The reported BLEU is 16.07, a 282% improvement over the optimized LSTM.

Methodological concerns The jump from 4.20 to 16.07 is the most important result in the manuscript. It demonstrates that multilingual pre-training is far more effective than custom LSTM engineering for this language pair. However, the comparison is not controlled. The improvement could be due to several factors:

- The Transformer architecture, which handles long-range dependencies better than LSTMs.
- The pre-training data, which includes hundreds of millions of sentences across 200 languages.
- The tokenization, which was learned on a much larger and more diverse corpus.
- The model size, which is more than ten times larger than the optimized LSTM.

To claim that Transformers are fundamentally superior, the authors should at least compare the NLLB result to a Transformer trained from scratch on the same 50K sentence pairs. Otherwise, the 4x improvement may be attributed primarily to scale and pre-training rather than architecture.

Benchmark context The manuscript does not compare the 16.07 result to published benchmarks. For context, the WAT 2019 Myanmar-English competition saw winning systems score substantially higher, but those systems were trained on task-specific data with back-translation and ensembling. A zero-shot NLLB score in the mid-teens is plausible for a generic test set. If the test set is from the same domain as NLLB's training data, 16.07 may even be on the low side, suggesting tokenization or encoding issues. The manuscript should state the test set source and domain.

Missing details The manuscript does not specify:

- The source and target language codes used for the tokenizer.
- The decoding settings: beam size, length penalty, maximum length.
- Whether the output was detokenized and lowercased consistently with the references.
- The test set size and whether multiple reference translations were available.

Recommendations

- Document the exact inference settings and tokenizer behavior.
- Report the test set size, domain, and tokenization.
- Compare the result against published NLLB-200 scores on FLORES-200 or WAT benchmarks.
- Show sample zero-shot translations to illustrate quality and failure modes.

1.4 Phase 4: NLLB-200 + LoRA Fine-Tuning (BLEU 21.87)

Experimental design The authors fine-tune NLLB-200 using Low-Rank Adaptation (LoRA) to avoid the memory cost of full fine-tuning. The reported BLEU after one epoch is 21.87, an improvement of 5.8 points (+36%) over the zero-shot baseline.

Methodological concerns LoRA is an appropriate choice for this hardware constraint. The parameter reduction claim (600M to 2-5M trainable parameters) is technically correct. However, the manuscript reports only one epoch of fine-tuning. One epoch is not enough to establish that the model has converged. Many fine-tuning runs continue to improve for several epochs. The reported 21.87 may be an underestimate or an unstable early result. The manuscript should report validation BLEU for at least five epochs.

The manuscript also does not specify the LoRA hyperparameters. Critical details missing include:

- Rank r and alpha scaling.
- Which modules were adapted (q_proj, v_proj, k_proj, o_proj, or others).
- LoRA dropout.
- Learning rate and optimizer.
- Batch size and gradient accumulation.
- Whether the learning rate was warmed up or decayed.

The standard Hugging Face PEFT configuration for NLLB-600M is often r=8, alpha=32, target_modules=["q_proj", "v_proj"]. If the authors used a different configuration, it should be documented.

The claim that LoRA achieves 95% of full fine-tuning quality is unsupported. Without a full fine-tuning baseline, this is an estimate rather than an empirical result.

Burmese-specific considerations LoRA adapts the model to the target data distribution. For Burmese, the most important question is whether the model is learning Burmese-specific patterns. A higher rank (r=16 or r=32) may capture more script-specific nuances than r=8. The manuscript should test this. Additionally, including the feed-forward (MLP) layers in the LoRA target modules may improve domain adaptation.

Recommendations

- Report validation BLEU for 5-8 epochs to show convergence.
- Document all LoRA hyperparameters precisely.
- Compare LoRA ranks r=8, r=16, and r=32.
- Consider including MLP layers in the LoRA target modules.
- If hardware permits, include a small full fine-tuning baseline to validate the 95% quality claim.

1.5 Phase 5: Data Augmentation with OPUS-100 (Expected BLEU 25-30+)

Experimental design The authors plan to combine the original 50,000 local sentence pairs with approximately 103,000 Burmese-English pairs from OPUS-100 and train for three additional epochs. The expected BLEU target is 25-30+.

Methodological concerns This section is entirely speculative because no experiments are reported. The projection of 25-30+ is optimistic for several reasons.

First, OPUS-100 is a noisy, English-centric web corpus. It is sampled from the broader OPUS collection and may contain misaligned sentences, mixed domains, and low-quality translations. For Burmese-English, it may also contain Zawgyi-encoded text or inconsistent normalization. The manuscript must describe how the OPUS-100 subset was filtered and whether it was checked for overlap with the local test set.

Second, simply doubling the training data does not guarantee a 3-8 BLEU point improvement. In low-resource MT, diminishing returns are common. The relationship between data size and BLEU follows a power law, not a linear trend. Without experiments, the 25-30+ target is not substantiated.

Third, the most reliable method for moving from the low-20s to the high-20s BLEU in low-resource MT is back-translation. The procedure is:

1. Take a large monolingual English corpus.
2. Translate it into Burmese using the current best model.
3. Use the synthetic Burmese-English pairs as additional training data.
4. Fine-tune the model on the combined real and synthetic data.

This technique has been shown to add several BLEU points for Myanmar-English systems and is absent from the manuscript's plan.

Recommendations

- Treat Phase 5 as a planned experiment with a hypothesis, not as a projected result.
- Filter OPUS-100 for quality: remove low-confidence alignments, deduplicate pairs, normalize encodings, and exclude extreme length ratios.
- Add back-translation from monolingual English data to the augmentation plan.
- Report intermediate results after each epoch of training on the augmented data.
- Use the combined dataset to test the Phase 4 LoRA setup for 5-8 epochs, not just three.

2. Evaluation and Metrics

BLEU limitations for Burmese The manuscript uses BLEU as the only evaluation metric. For Burmese, this is problematic because BLEU is sensitive to tokenization. Burmese has no word boundaries, so the tokenizer strongly influences the score. A translation that is semantically correct but segmented differently from the reference will receive a low BLEU score. The manuscript should report chrF++ (character n-gram F-score) alongside BLEU, as chrF++ is more robust to segmentation variation.

Statistical validation All reported BLEU scores are single point estimates. The manuscript does not provide confidence intervals or significance tests. For low-resource test sets, a difference of 1-2 BLEU points can be due to chance. Bootstrap resampling should be used to report 95% confidence intervals, and phase-to-phase comparisons should be tested for statistical significance.

Human evaluation The manuscript does not include human evaluation. Even a small-scale study (50-100 sentences) with fluency and adequacy ratings would strengthen the claims about translation quality.

Recommendations

- Report BLEU, chrF++, and TER for all experimental phases.
- Provide 95% confidence intervals via bootstrap resampling.
- Conduct a small human evaluation for fluency and adequacy.
- Use a standard benchmark such as FLORES-200 or a WAT test set to enable comparison with published work.

3. Reproducibility

The manuscript lists hyperparameters, dependencies, and project structure, but several elements needed for reproduction are missing:

- Random seeds are not provided.
- Exact train/validation/test split sizes and methodology are not stated.
- The preprocessing pipeline is not described in enough detail.
- Code and data availability statements are absent.
- No training curves, logs, or sample outputs are included.

Recommendations

- Add a reproducibility appendix with random seeds, data splits, BPE training details, preprocessing commands, and exact software versions.
- Provide a requirements.txt file with pinned dependency versions.
- Include sample translations and error analysis examples.

4. Recommendation

Major Revision Required

The manuscript follows a sensible trajectory and the reported numbers are plausible. However, it currently reads as a project report rather than a research contribution. The authors should revise the work to include:

1. A detailed Burmese preprocessing and normalization analysis with data quality metrics.
2. Controlled ablation studies for Phase 2 and LoRA hyperparameter experiments for Phase 4.
3. Multiple evaluation metrics with bootstrap confidence intervals.
4. Convergence analysis for the LoRA fine-tuning.
5. A realistic, experimentally grounded Phase 5 plan with data filtering and back-translation.
6. A reproducibility appendix with seeds, splits, code, and preprocessing details.

If these revisions are addressed, the work could become a useful empirical case study in low-resource machine translation for Burmese-English.